Building the Synoptic Key of Life website

An Independent Study

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Abstract

SKOL is a synoptic key. It helps amateur mycologists navigate the open access taxonomic literature covering fungi. It guides the citizen scientist through the process of creating technical descriptions of fungal collections. It complements photography-based sites such as Mushroom Observer, and iNaturalist. It increases the value of the other records that may be taken about a collection for a single species, including DNA barcodes, and Fungarium accessions.

SKOL is in alpha test at https://synoptickeyof.life/skol as of mid February 2026 and should progress to beta test by May 2026.

1. Background and motivation

Many are familiar with nature field guides that have a key which presents a series of couplets with questions like “if your bird has brown feathers, go to couplet 7; otherwise continue with couplet 2.”. See Figure 1 for a full example from Michael Kuo. This is called a dichotomous key. It takes some practice to use a dichotomous key, as you need to be able to backtrack to choices that you are uncertain of or which you found ambiguous.

There are many excellent dichotomous keys to fungi such as the keys produced by the Pacific Northwest Key Council @pnwkeycouncil. Another excellent key that uses couplets is at Michael Kuo’s mushroomexpert.com @kuo2025.

A synoptic key allows a taxonomist to start with any feature that stands out to them, and then helps them narrow their search by making additional observations. There are paper synoptic keys, but computers open up the possibilities for synoptic keys significantly. An excellent synoptic key to fungi is the commercial product MycoKey @mycokey2020.

All of these keys are hand-built and thus require more hand work to keep them up-to-date.

Key to 9 Species of Armillaria in North America

1. Stem without a ring or cobweb-like ring zone.	*Armillaria tabescens*
1. Stem with a ring or ring zone.	2
2. Mature cap with prominent scales.	3
2. Mature cap smooth, with scattered fine hairs, or with tiny scales, but without prominent scales.	5
3. Found on hardwoods; distributed from the Appalachian Mountains eastward; rare; benign.	*Armillaria gemina* (see *Armillaria solidipes*)
3. Found primarily on conifers; widely distributed in northern North America; common; parasitic.	4
4. Found at high elevations in interior forests with fir; stem base swollen.	*Armillaria altimontana* (see *Armillaria solidipes*)
4. Found in a variety of coniferous forests; stem bases equal or narrowed.	*Armillaria solidipes*
5. Cap color orange-brown; stem dark and colored like the cap; growing on hardwoods (especially species of *Alnus*) in western North America.	*Armillaria nabsnona*
5. Not completely as above.	6
6. Stem with a thick, membranous ring; stem bases pointed due to growth habit in dense clusters; basidia *without* basal clamps.	*Armillaria mellea*
6. Ring ephemeral or cobwebby; stem bases not pointed; (at least some) basidia basally clamped.	7
7. Pathogenic and parasitic, invading the wood of hardwoods (in the northeast) and conifers (in the Pacific Northwest) with black rhizomorphs; mushrooms typically fruiting from visible wood.	*Armillaria sinapina*
7. Harmlessly saprobic on hardwood (and occasionally conifer) roots east of the Rocky Mountains; not displaying prominent black rhizomorphs in the host wood; mushrooms often appearing terrestrial, though occasionally fruiting from visible wood.	8
8. Mating in the laboratory with previously identified isolates of *Armillaria gallica*; widely distributed east of the Rocky Mountains.	*Armillaria gallica*
8. Mating in the laboratory with previously identified isolates of *Armillaria calvescens*; northeastern in distribution.	*Armillaria calvescens* (see *Armillaria gallica*)

Fig 1. Michael Kuo's Key to 9 Species of Armillaria in North America
SKOL seeks to automatically build a synoptic key to fungi that keeps up with the literature as papers are published. It uses as little human judgement as possible, leveraging several AI techniques to expose the open literature. Developing familiarity with the published literature requires years of effort, usually more than an amateur may have the bandwidth for.

In recent years, it has gotten difficult to fund the meticulous descriptions needed for traditional morphological taxonomy, @Agnarsson2007. A few have suggested abandoning traditional taxonomy entirely in favor of DNA-based clades @tautz2003plea. But this risks disconnecting DNA entirely from morphology @DeSalle2005. DNA-based cladistics provide a solid basis for evolutionary relatedness. The natural focus of DNA analysis of fungi is the proteins coded for in the DNA.

An excellent example of DNA-based work that ties back to traditional morphology is @Hibbett1994, in which they track down the single allele that causes some forms of *Lentinus tigrinus* to form a partial veil covering its gills that does not open up into a ring.

There is a growing community of amateur mycologists who are doing large-scale DNA fingerprinting of fungal collections. DNA fingerprinting sequences a standard part of an organism's genome. The mycological community has chosen ribosomal DNA @Vilgalys1990RapidGI Internal Transcribed Spacer (ITS) @schoch_nuclear_2012, for species and Long Subunit (LSU) @kurtzman1998identification for genus. There are a few dedicated amateurs such as Rod Tullos @tulloss2026 who have taken up the task of morphological description of new and existing species.

SKOL seeks to promote the practice of creating detailed morphological collections. These descriptions can be made part of the informal literature. If collections are then associated with DNA sequences in GenBank, pictures in MO or iNaturalist, or specimens in fungaria, the value goes even higher.

2. Organizing principles

SKOL is heavily influenced by the Open Source community, and projects such as WIkipedia. I have observed that my Open Source contributions are the ones that have survived the longest. Most work that I've done commercially stopped being useful not long after I left the company, often even before I left.

To this end, all the data we use is either public domain or published open access. The source code is open source, and all essential components are open source. It's OK to use proprietary tools (like neo4j) to diagnose data issues or otherwise explore the project (like Claude Code, @claude2026code, or Trello @trello2026).

My goal is to create an active and vibrant community around SKOL. I am consciously applying the founder of Wikipedia's @Wales2025SevenRules Seven Rules of Trust:

1. Make It Personal

   People may choose their own id on the site. This is only important to a small fraction of the general population, but I'm part of the group that cares deeply about my electronic identity.

   People who want to use SKOL for publications can embargo their descriptions for up to a year. This is to encourage people who want the descriptions for publications to use the tool.

2. Be Positive About People

   I'm emphatic about giving credit to the many people who have helped me on this project. People should know that their contributions are valued.

3. Create a Clear Purpose

   SKOL is a Synoptic Key. We support the creation of technical taxonomic descriptions.

4. Be Trusting

   Anyone can use the web site anonymously to search for results. If they want to save their descriptions, they need only provide an email address. I'm applying as light a touch as I can in dealing with spam, relying on detection after-the-fact rather than putting large barriers up.

5. Be Civil

   We assume good intent in all interactions.

6. Be Independent

   At the start of the project, I am able to self-fund all needs. As project needs grow, I plan to follow the Public Radio model of funding, possibly through Patreon.

7. Be Transparent

Transparency to the public is a core requirement. How the project is managed is as open as practical, e.g. the source code is on GitHub and planning is in a public Trello. Processes are documented as they are created.

Almost all actions on the web site leave permanent public cookie crumbs.

3. Models

The project uses a range of AI and Machine Learning methods to solve separate problems.

Classifier

My original work in 2019 involved labeling a couple hundred issues from three journals (@mycologia_journal, @mycotaxon, @persoonia) to extract Nomenclature, Description, and Misc-exposition blocks. I explored about half a dozen classification models and settled on logistic regression.

I recently put about a week into exploring bidirectional RNNs for classification, but they are expensive to compute and do not perform as well as logistic regression.

Sentence Embedding

Building on the Dr. Drafts SOTA Literature Search (Gisolfi2024), Shintaro Osuga, Chris Murphy, and I created dr-drafts-mycosearch, which embeds descriptions into a searchable semantic space with SBERT, allowing fast similarity testing for new descriptions. SBERT allows precomputation of fixed-length semantic embeddings of whole descriptions which can be easily searched with cosine distance.

Since the embeddings are semantic, there is a lot of flexibility in input terminology. Users can use non-technical terms to get reasonable results. In principle, the embeddings should even be language independent, opening up literature in languages-other-than-English (LOTE). I have not run systematic experiments, but would observe that I've gotten a few Latin and German results while testing. LOTE has not been in scope for the independent study.

Taxon to JSON

Padmaja Kurumaddali, Patrick Le, and I use the mistral small language model (SLM) to create JSON structures representing descriptions semi-formally. We applied fine tuning to mistral in an effort to improve the quality of the translations. We achieved better results by prompt engineering.

More recently, I used Claude Code to identify several more areas for improvement.

The biggest difference comes from constrained decoding with the outlines library @outlines. I created a schema that constrains the output to JSON expressions 2-6 levels deep.

I explored injecting PATO (Phenotypic Quality Ontology) and FAO (Fungal Anatomy Ontology) context into prompts, guiding the model to use standardized scientific terminology rather than inventing ad-hoc terms. The ontologies proved too sparse to be of general use. The injection process generates intermediate layers of JSON nesting of limited utility like “feature”.

Vocabulary Normalization has been helpful for standardizing the terms in the JSON structures, by post-processing extracted terms by mapping them to standardized ontology terms using similarity matching with a configurable threshold.

When encountering novel terms not in ontologies, Graceful Degradation preserves original terms with confidence annotations rather than forcing incorrect mappings. It categorizes matches as HIGH/MEDIUM/LOW/NOVEL.

Direct JSON feature selection

SKOL implements a feature selection tool offering searchable nested menus based on successive layers in the JSON descriptions. The first menu lists all the top-level terms. The second menu is created automatically with all the second-level terms in the JSON descriptions that occur below the selected top-level term. Menus continue all the way down to the leaves.

The resulting menus are challenging to use. The top level menu is huge with a lot of similar technical terms and some noise entries like “`` `& checking` ``”, or raw metrics without context like “`15-18 µm`”.

Decision tree feature selection

SKOL implements two feature selection tools based on ad-hoc decision trees. Behind the scenes, a search request pulls a larger number of results than those displayed to the user and builds decision trees for those descriptions.

One decision tree works with the descriptions as text, providing terms which are good at discriminating among the candidate taxa. This has been moderately successful, though it tends to produce a lot of taxonomically uninteresting suggestions such as “842 in” or “low”. Though it provides some interesting hints such as “the stipe”.

The other decision tree works with the JSON descriptions. The JSON descriptions are flattened to stacks of terms leading to a leaf and then placed in a decision tree of all the matching descriptions. This consistently produces taxonomically interesting feature suggestions.

4. Data quality

The Nomenclature and Description blocks generated by the classifier are very good for the original journals they were created from.

The models have not generalized well to other journals. Older journals have been especially poor, with noisy OCR and different typographical conventions. Many older journals put the nomenclature at the start of a description paragraph in italics. The nomenclature doesn't necessarily carry all the authorship indications common in modern publications.

Newer publications have consistent headers which should make it easier to extract nomenclatures and descriptions, but the training data is on articles that are old enough to not have all of these conventions.

There is a clear need for more manual labeling, which I have explicitly left out of scope for the independent study, but this may be necessary to really make SKOL useful to real practitioners.

I suspect that better Description blocks would improve the JSON structures, at least by eliminating some of the noisier top-level keys.

5. User Interface Design

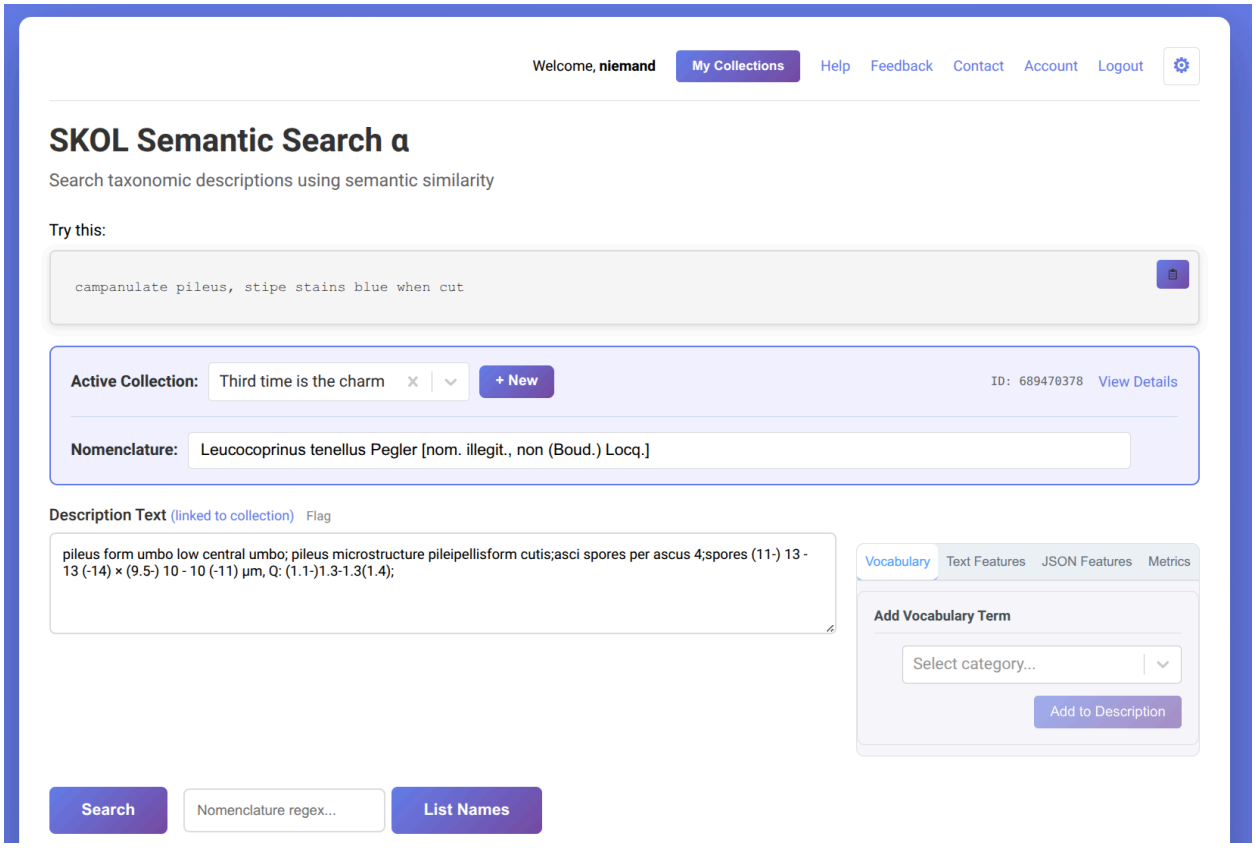


Fig 2. The Description Search page

The Description Text is the central feature of the UI (see Fig 2.). This is where the taxonomist builds a technical description of the organism they have in front of them. Everything else on the page supports that activity.
The Search button allows a regular expression to apply to
The Active Collection area allows the taxonomist to organize all their collections separately. The Nomenclature field holds the current guess for the identity of the collection, generally a species name.

Above the Active Collection is a sample description (Try this) that during the alpha phase of the project, taxonomists can use to “kick the tires” of the system.I used this extensively in early development just to have a legitimate description to search with.

The Feature Tools

To the right of the Description Text, we find the feature suggestion tools. These are tools for identifying features to add to the description.

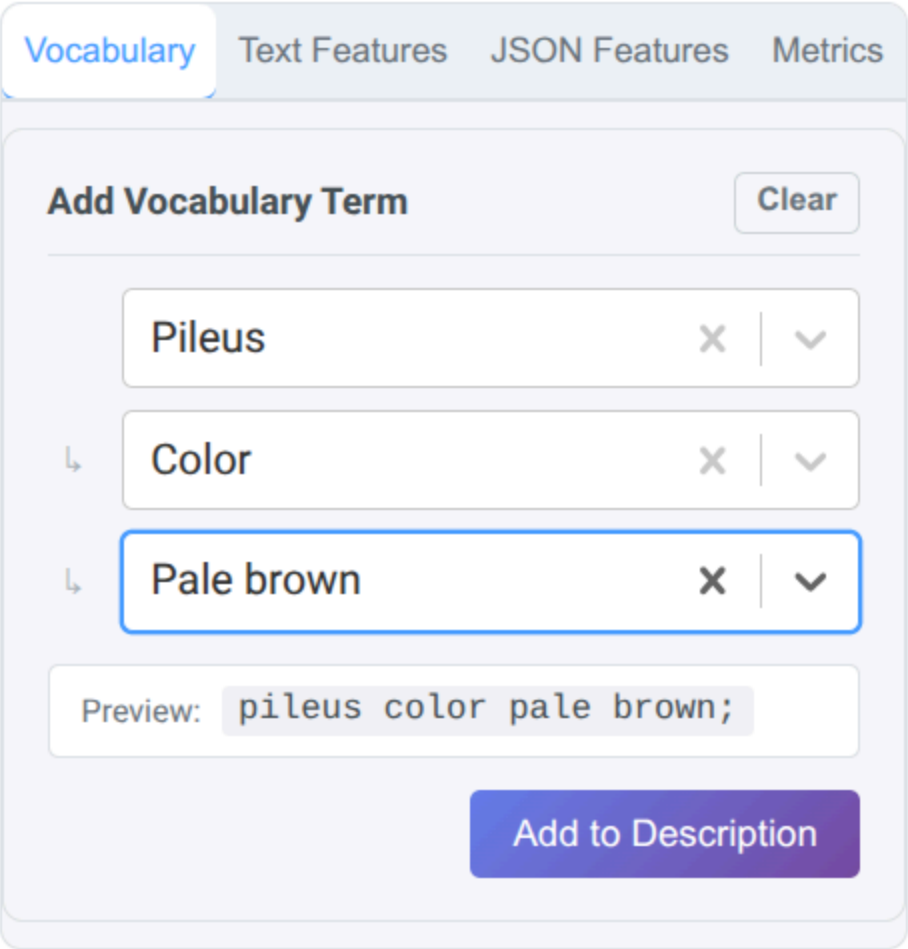


Fig 3. The Vocabulary tool

The Add Vocabulary Term tool provides a list of nested menus which allow the taxonomist to select from successive layers of the JSON description hierarchy. This is a static set of menus which is not based on the current description.

The Text and JSON Features tools are both based on decision trees for the closest matches to the current description.

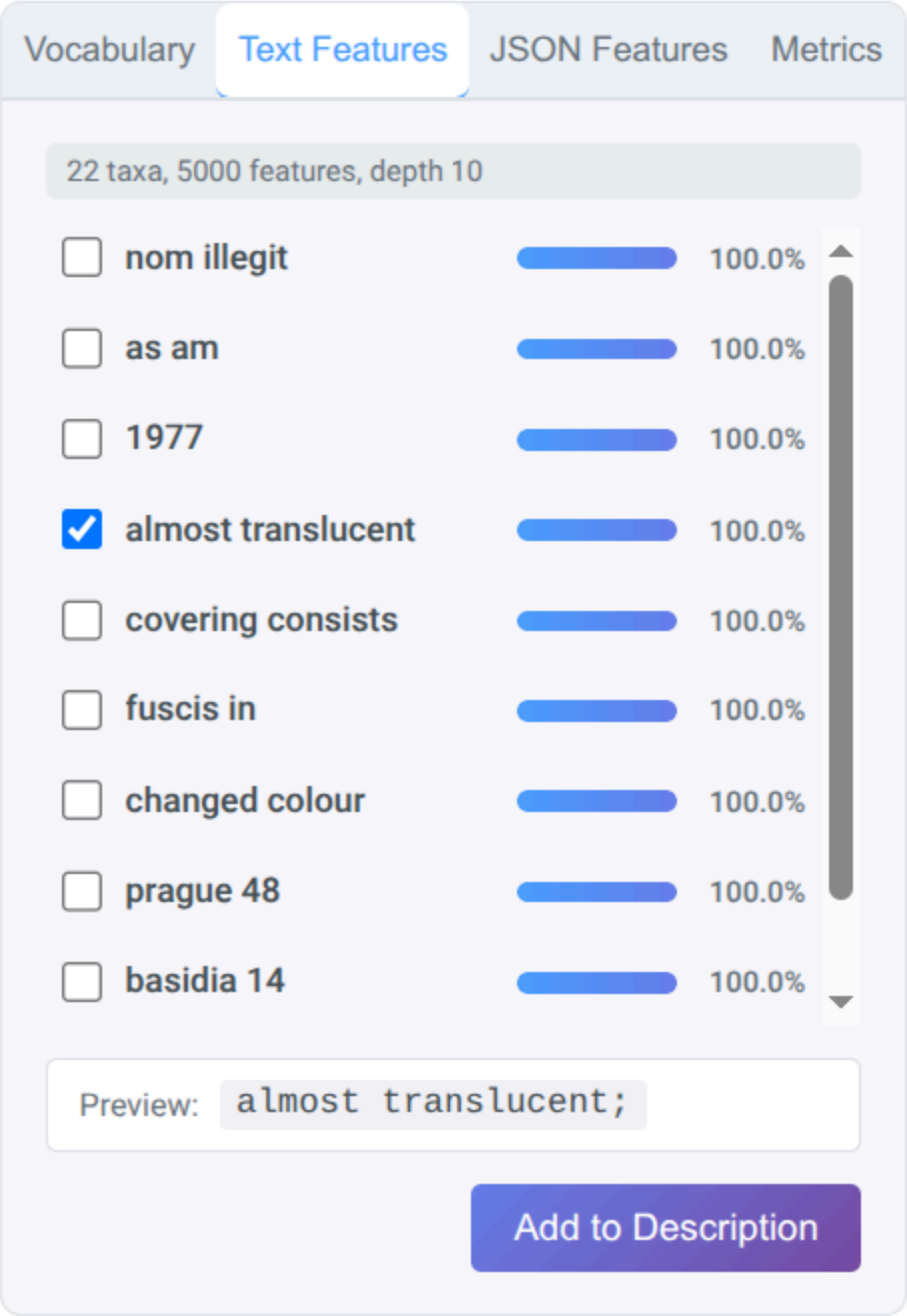


Fig 4. The Text Features tool

The Text Features tool shows which vocabulary best separates the top few matching descriptions. It's not always clear that the vocabulary differences are taxonomically meaningful.

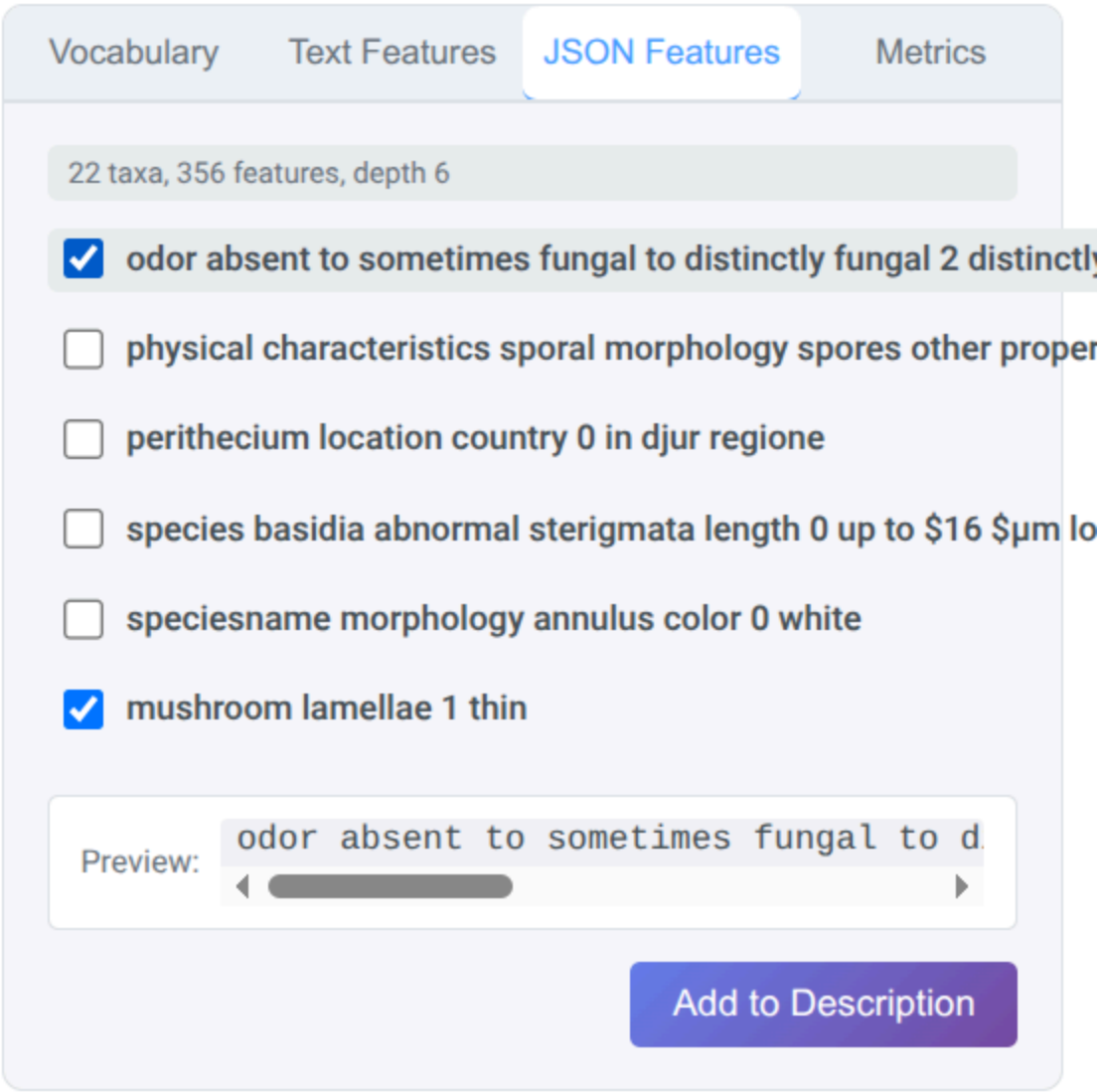


Fig 5. The JSON Features tool

The JSON Features tool shows the vertical column of keys to each leaf which best separates the top matches. I personally think this is the best of the 3 tools. It provides whole phrases that can be easily edited to apply to the current collection.

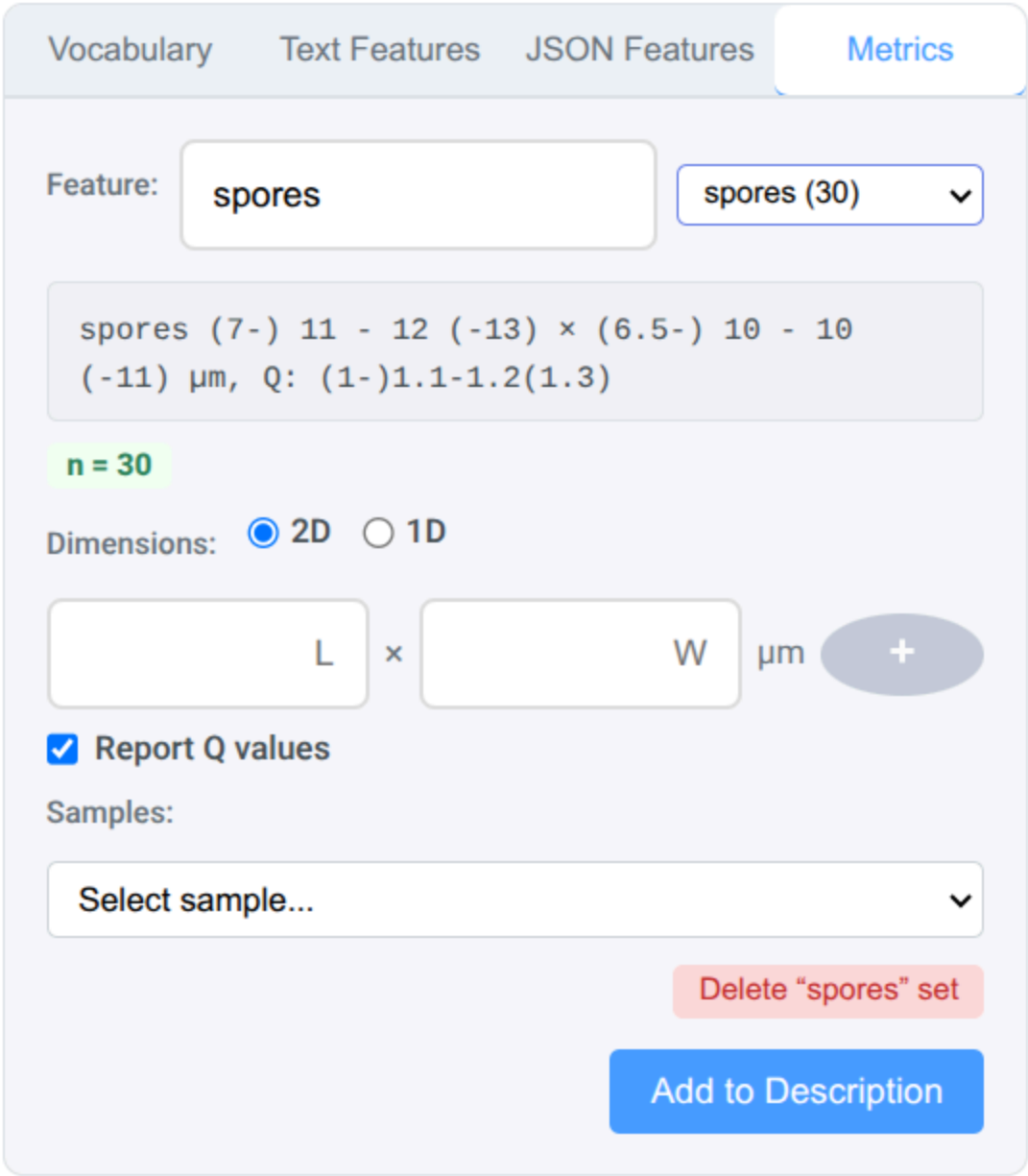


Fig 6. The Metrics tool

The Metrics tool allows the Taxonomist to record measurements of microscopic features. The user can record statistics about any feature they care to name, either 1 dimensional or 2 dimensional. It defaults to spores. The measurements are permanently associated with the collection. The statistics are (min)-1st quartile-3rd quartile-(max). 2D features can also report Q, the ratio of length to width, where 1.0 is round and larger numbers are more and more elongated.

External Identifiers

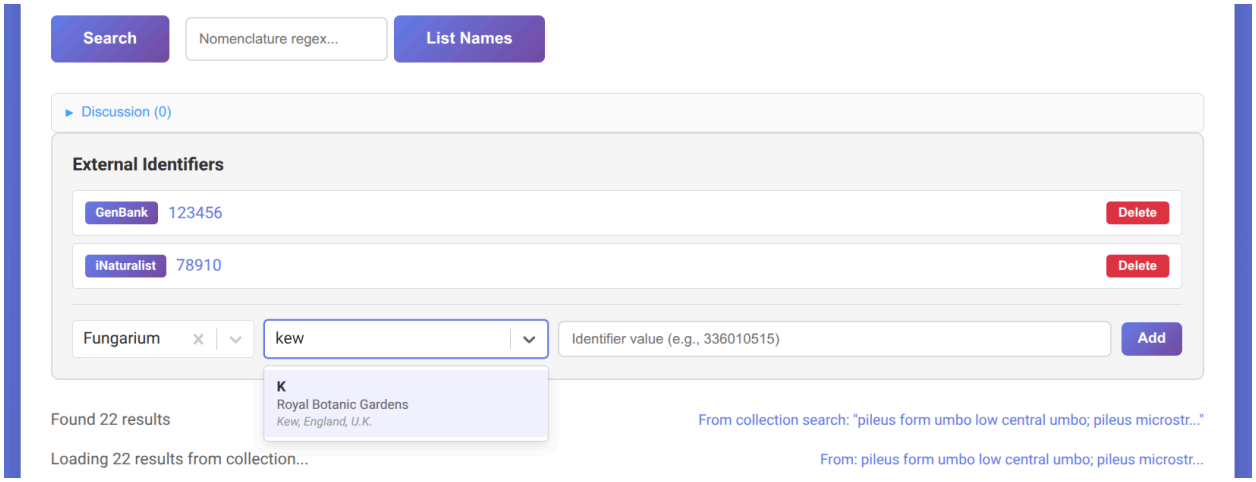


Fig 6. External Identifiers

Below the Search button we have External identifiers, which allows the taxonomist to link to external records for the collection, such as iNat or Mushroom Observer record, DNA information, or a link to the herbarium.

This list of Fungaria is based on the list from the New York Botanical Garden with a facility for making local amendments, though I've not had to apply any yet.

Coming soon is a button to copy the Nomenclature and Description to an iNaturalist observation.

The Search Results

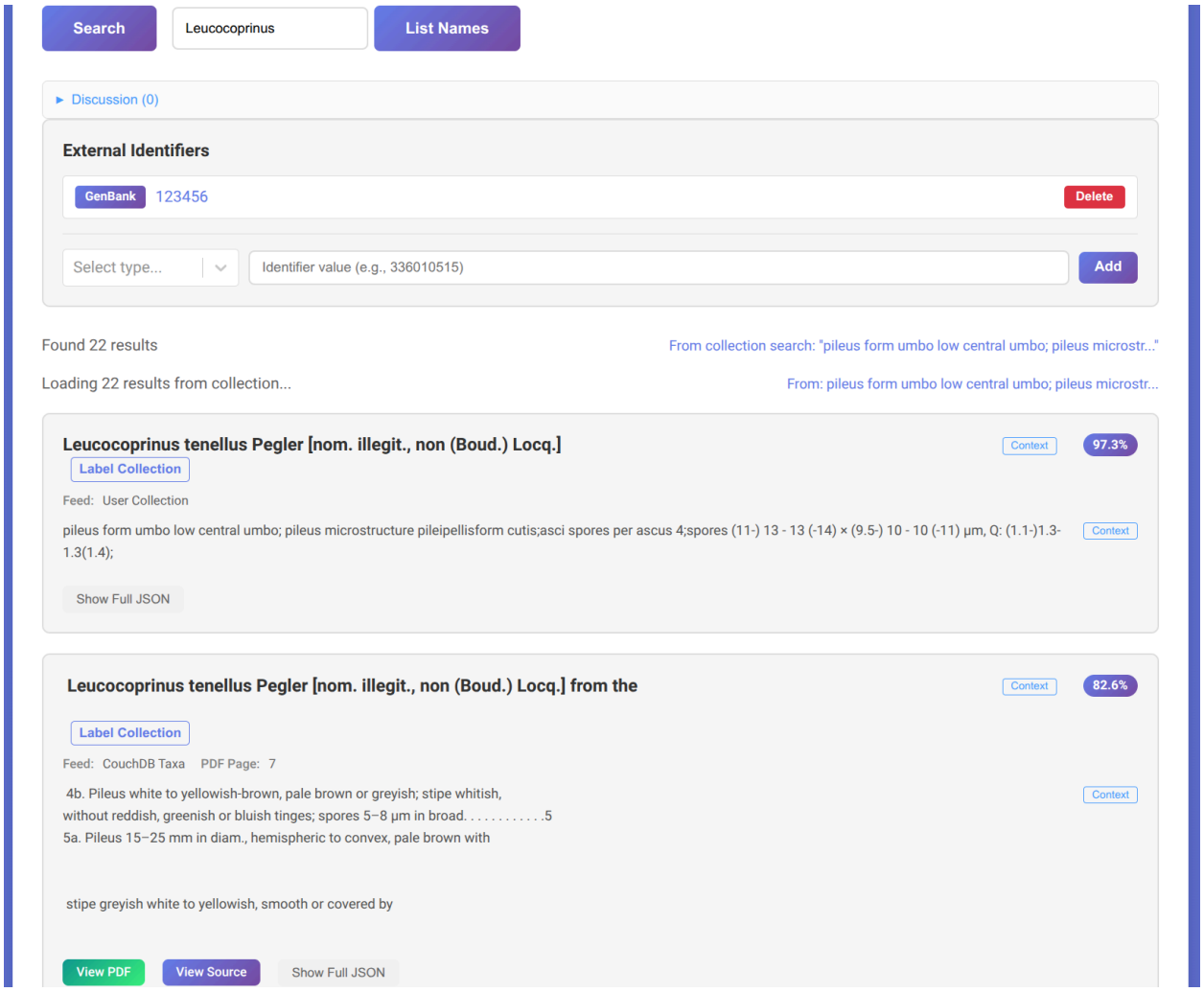
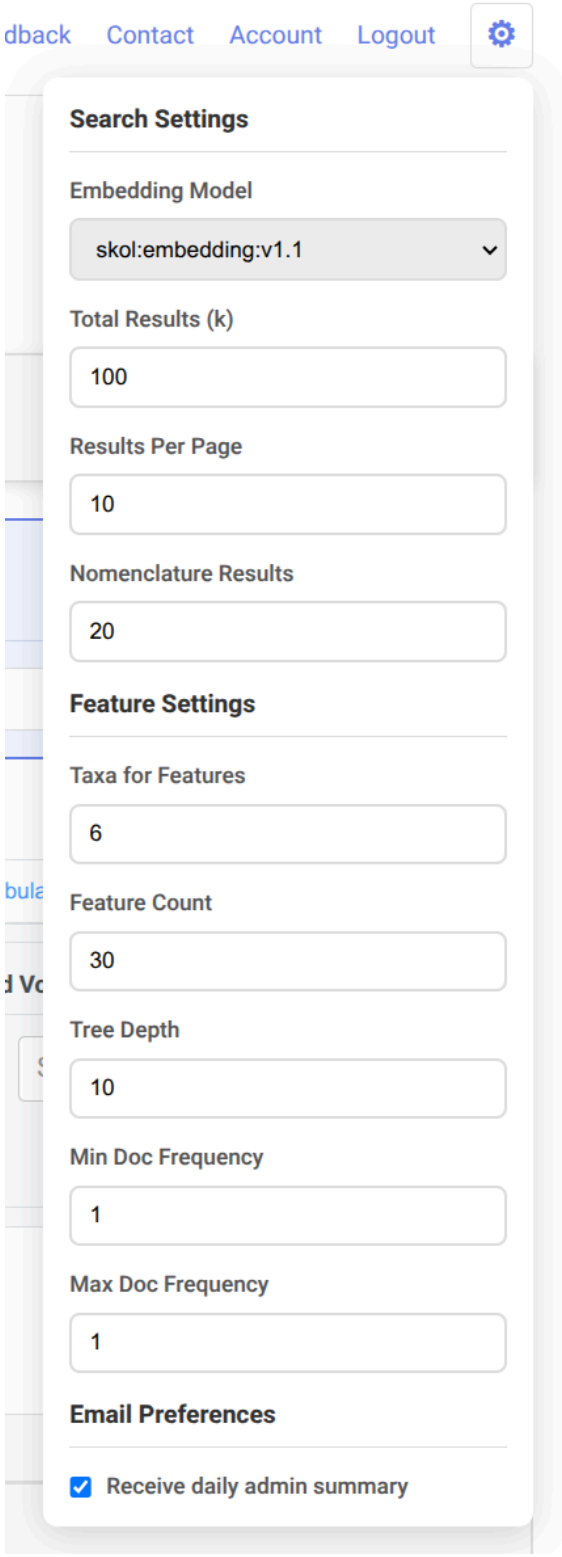


Fig 7. The Search results

Next we have search results. Each search result has a Nomenclature (meant to be a taxon name) and a Description. Both fields have a Context button that shows the original text surrounding the matches. This has proven valuable for data debugging.

Each result also has View PDF, View Source, and Show full JSON. View PDF brings up a PDF reader with the local copy of the whole document. If possible, the local copy will be paged to the exact page that the Nomenclature comes from. View Source takes you to the human-readable page at the publisher for the Nomenclature. Show full JSON is a debugging tool that gives the raw result record from dr-drafts-mycosearch. If the buttons are missing it is generally an indicator of a record that is no longer valid due to recalculation of taxa.

Settings



In the upper right is a gear icon that holds the taxonomist's default settings like the number of results to show, the number of records to use in the decision tree, and some hyperparameters for the decision tree.

The embedding model is the search space. The embeddings labeled `skol:embedding:*` include all indexed sources. Additional models are planned, such as a keys-only embedding that only brings up dichotomous keys.

Taxa for Features controls the number of taxa in the decision trees for the two decision tree-based feature tools. The other parameters allow tuning the tree model.

Fig 8. Settings

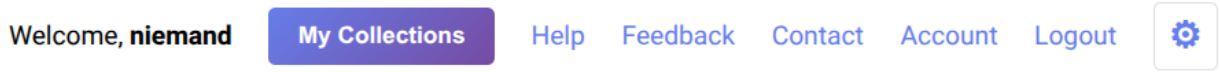


Fig 9. Top menus

Across the top, or for mobile in a hamburger menu in the upper left, we have common operations. Every page in the site has a Feedback button that allows the taxonomist to provide feedback about the given page. If the taxonomist has registered GitHub credentials, the feedback will be recorded as a GitHub issue on the `piggyatbaqaqi/skol` project. On the main page we have a Contact button for web contacts.

Collection Discussion Threads

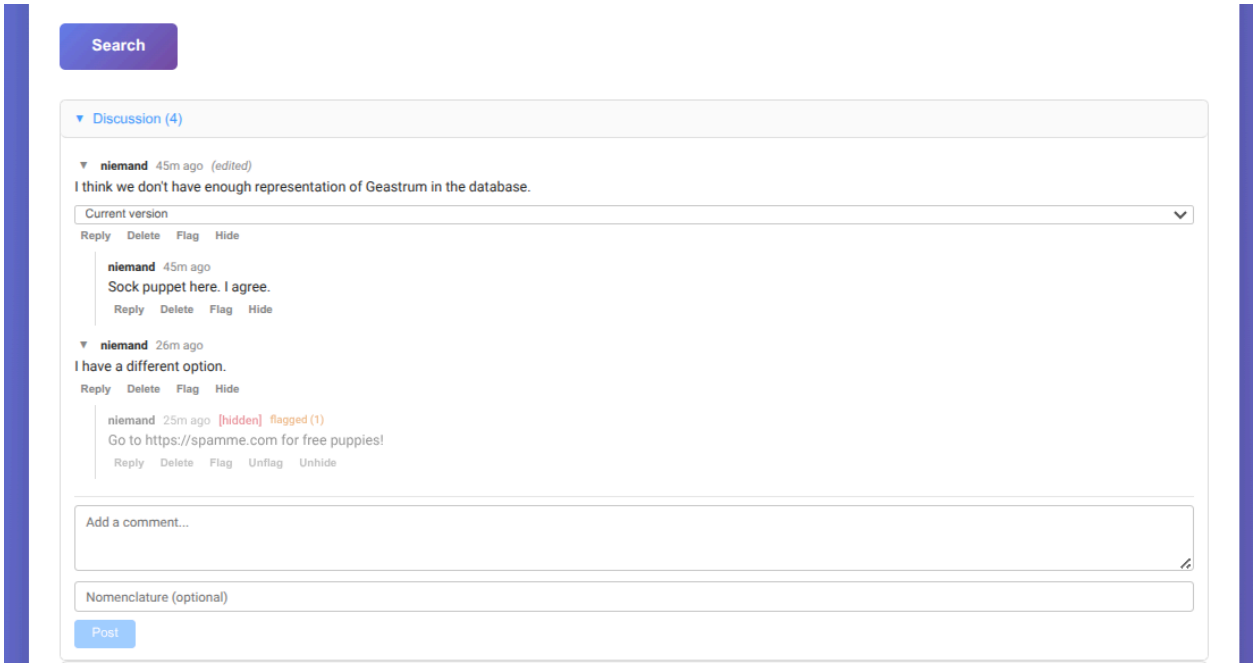


Fig 10. Discussion Threads

Each collection has a Discussion section. It provides a simple nested thread model for comments. Users can hide their own posts. Anyone can flag a comment as inappropriate. Staff and superusers can hide any post or undo a flagged label.

This capability is the start of a community building capability.

Mobile

I’ve been demonstrating and testing the web site from Android, so I’ve added a number of features which make things better on mobile.

The hamburger menu in the upper left of the mobile site improves navigability of the site.

As the databases are very much in flux, I've added a semi-automated refresh for some of the generated databases. If we try to access a database and it is not present, we launch an automatic rebuild of it.

6. Project tracking and planning

The project is organized using a very light version of the XP Planning Game, run through Trello.

The XP Planning Game @beck1999extreme is run as a series of Iterations of 1 week, where the amount of work achieved in the previous iteration is used to decide how much work to schedule for the next iteration. Work is estimated in abstract complexity rather than in time so that overhead and velocity can be measured directly.

The unit of complexity for this project is "Create a scraper for a new web site with Claude Code.". All other stories and tasks are estimated as multiples in complexity of the unit task.

The planning units also allow us to manage the backlog of stories and tasks to assure that it fits in the available time.

7. Productionization

The Redis, Couchdb, and Neo4j databases are all run in docker containers based on configuration created by Dr. Michael Fudge for IST769. Storage has been moved outside of the docker ecosystem to directories in the /data/skol filesystem, which can be more easily backed up.

Packaging

The code is bundled into three debian packages, `skol_*_all.deb`, `skol-django_*_all.deb`, and `dr-drafts-mycosearch_*_all.deb`. The two skol packages are installed into build-version-specific directories which are then made active by switching symbolic links. This allows for minimal system downtime during upgrades, and for easy rollback.

All three packages use *`major.minor.subminor-build`* for their version numbers. The build number is automatically incremented when the deb file is created. The installation directories all live under `/opt/skol`.

The deb packages are copied automatically with

```
(base) piggy@puchpuchobs:~/src/github.com/piggyatbaqaqi$ watch -c "rsync
-av dr-drafts-mycosearch/deb_dist/dr-drafts-mycosearch_*_all.deb
skol/deb_dist/skol_*_all.deb skol/django/deb_dist/skol-django_*_all.deb
skol:"
```

Installation is handled with the `watch_incremental` command built just for this project:

```
root@skol:/home/piggy# /opt/skol/bin/watch_incremental
--postinstall="systemctl restart skol-django" './skol_*_all.deb'
'./skol-django_*_all.deb' './dr-drafts-mycosearch_*_all.deb'
```

All credentials are in `/home/skol/.skol_env`. Every other point in the system that needs credentials gets them from there. In the future, I'd like to use `gpg-agent` to hold passwords only in memory. It will require intervention by a human at boot time.

Email

I configured the `synoptickeyof.life` domain as a modern email source and recipient. In addition to setting up an MTA (Mail Transport Agent) and MX (Mail Exchanger) records in DNS, there are modern email safety requirements. These are SPF, DKIM, and DMARC records.

SPF (Sender Policy Framework) is a DNS record which tells hosts on the Internet which hosts are authorized to originate email for `synoptickeyof.life`. This prevents spoofing.

DKIM (Domain Keys Identified Mail) provides the public key for other hosts to confirm that a given piece of email was legitimately generated by `synoptickeyof.life`. This makes spoofing and phishing more difficult.

DMARC (Domain-based Message Authentication Reporting and Conformance) tells email hosts on the Internet what to do with SPF and DKIM failures. It includes a special email address that can collect failed email messages to generate a report. I'm using the postmarkapp.com DMARC service for generating the weekly report.

Backups

Backups are done with cron jobs that do daily `rsync` commands to an attached external drive on `skol.synoptickeyof.life`. I have a second machine offsite which needs

new drives and a GPU which will act as a hot spare for the domain and allow offsite backups.

8. Comparison with aspirations

The Aspirations document describes behaviors I'd like the system to eventually exhibit. It classifies the behaviors into Minimal Viable Product, Planned for IST690, Stretch Goals, Nice to Have, and Research Projects.

The behaviors cover many aspects of the system ranging from user-visible behaviors, to social engineering components, to operational requirements. The priorities I set in Trello have often been things I did not think of for the Aspirations document. Because of this, I haven't followed a strict prioritization of the behaviors, other than the MVP.

I organically solved two of the Research Projects: Feature selection via decision tree, and REST authentication. I have to credit Claude Code for doing a fairly good job of setting up REST authentication while implementing other features, requiring only a few nudges. Feature selection via decision tree ended up being fairly straight-forward.

There are 6 minimal-viable-product behaviors in the Aspirations document (codes MI and MD). All 6 are complete.

There are 19 behaviors in the Aspirations document marked for completion as part of IST690 (code I). As of `Mar 1, 2026` 18 are complete. One has been dropped as too expensive.

There are 9 stretch behaviors in the Aspirations document (code S). 7 are complete, The other 2 are in IST690 Stretch Goals.

There are 8 nice-to-have behaviors (code N). 2 are complete. The other 6 are in IST690 Stretch Goals.

There are 8 Research projects (code R). 2 are complete. 1 is redundant with a stretch goal. 1 is in the IST690 Stretch Goals. The rest are in the plain Backlog.

9. What I learned about UX

The advent of Claude Code radically changed what I learned from what I anticipated learning. I thought I would develop technical expertise with django, react, and related technologies. Instead, I learned how to work at a much higher level of abstraction. I find that I can describe the UX goals I have and get reasonable implementations automatically.

It is still important to know what the tools can do, but knowing how to make them do what they do is a lot less critical than it was just 2 years ago.

10. The future

I plan to continue this project for many years to come. I expect as we move through alpha and beta testing we'll find a lot of issues. I find that I'm adding backlog items almost as fast as I'm retiring them. It is interesting to observe that the backlog lists (Backlog, Anomalies backlog, IST690 Backlog, and IST690 Stretch Goals) total roughly the same as the lists that record what I've accomplished as part of IST690.

I don't anticipate being able to maintain the pace I've kept during the independent study class--I hope to have a day job again soon :-). I've kept this project going for almost 7 years and see no reason that I wouldn't go for at least another 7. I anticipate finding more people with the desire to paint this particular fence.

The immediate future will definitely include addressing some of the data quality issues. Hand labeling more documents is definitely in my future. Results from annotations of three contemporaneous journals have not generalized as well as I hoped.

I have thoughts about producing description templates that users could modify for particular morphologies, such as mushrooms with stems. It's easy to see how I could make these by hand, but in the spirit of the project they should be automatically derived from analyzing the literature. I don't really know how to do this, but I've been here before!

I may return to the finer-grained annotation I followed for the first few articles, splitting Misc-exposition into particular patterns like doi references, methods discussions, keys, and so-on. I anticipate that this will make Descriptions easier to identify.

I would like to expand the taxa covered. A subfield of Botany will be my next target, perhaps something like Bryophytes. I hope to attract developers who are interested in the work necessary to bootstrap new taxa.

11. Acknowledgements

I am grateful to my instructors and fellow students who have supported this project. I especially need to thank fellow student Chris Murphy for collaborating in both IST 718 and IST 664 for SKOL-related projects. JJ Balasi is also a treasured repeat collaborator for IST 664, and IST 736. My gratitude to additional collaborators Shintaro Osuga (IST 664), David Caspers (IST 718), Padmaja Kurumaddali & Patrick Le (IST 691). You all provided key insights for various aspects of SKOL.

Dr. Gregory Block provided me with an influential reading list. He introduced me to the tension between Tufte's data-ink ratio and cognitive load in a UI. His weekly collaboration made the Planning Game enjoyable and effective.

Annotated Bibliography

Jimmy Wales with Dan Gardner, 2025, The Seven Rules of Trust, Crown Currency @Wales2025SevenRules.

To say that this book is influential for me is an understatement.

This book is an easy read, full of detail and stories about building trust. Stories include Richard Nixon's early scandal about a secret fund, Airbnb's crisis when EJ's apartment was sacked, Buckley's Cough Medicine ("It tastes awful, And it works"), the University of Chicago's institutional neutrality, Richard Dawkins' story of a young scientist contradicting an established scientist, and many other stories.

Wales goes through his Seven Rules of trust, both as a general concept and specifically how Wikipedia embodies them.

This is the first time I've seen Frei & Morriss's Trust Triangle @Frei2020BeginWithTrust. Authenticity, empathy, and logic are the three core drivers of the Trust Triangle. This aligns very well with my personal development objectives.

Here are Wales's Seven Rules of Trust, plus a transcendent rule.

- "make it personal"
- be "positive"
- have "a clear purpose"
- "be trusting"
- keep it "civil"
- "be independent"
- "be clear"

And the rule that transcends them all:

- "It does exactly what it says on the tin."

I seek to apply all these rules to SKOL as outlined above under Organizing principles.

One of the first things I did upon finishing this book was to update SKOL's About page to demonstrate "a clear purpose". "Wikipedia is an encyclopedia." is the first of Wikipedia's "five pillars". So I wrote "SKOL is a Synoptic Key".

Here are all 5 pillars:

1. Wikipedia is an encyclopedia.
2. Wikipedia is written from a neutral point of view.
3. Wikipedia is free content that anyone can use, edit, and distribute.
4. Wikipedia's editors should treat each other with respect and civility.
5. Wikipedia has no firm rules.

I am tempted to incorporate these pillars directly into SKOL. Pillar 2 seems more specific to an encyclopedia than the others. Here are the 5 pillars of SKOL:

1. SKOL is a synoptic key.
2. SKOL descriptions are written in neutral technical language.
3. SKOL is free content that anyone can use, edit, and distribute.
4. SKOL's taxonomists should treat each other with respect and civility.
5. SKOL has no firm rules.

Don Norman, 2013, The Design of Everyday Things: Revised and Expanded Edition, MIT Press @Norman2013Design.

This highly accessible book discusses human-centered design. Norman asserts that usually "human error" stems from bad design. Bad design does not take human psychology into account.

This book introduced me to Affordances and Signifiers. Affordances are the possible interactions offered by a tool, and the Signifiers give cues to where and how to make the interaction possible. Many of the system behaviors I defined are Affordances, e.g. "It is possible to label a collection with a Nomenclature found by a search.". My challenge was to provide good signifiers, e.g. there is a button next to each Nomenclature result labeled "Copy" that exercises the Affordance.

Norman introduces seven stages of action: Goal, Plan, Specify, Perform, Perceive, Interpret, and Compare. Other than the goal, these are organized into three layers: Reflective (Plan & Compare), Behavioral (Specify & Interpret), and Visceral (Perform & Perceive). A failure in any of these stages makes it difficult to achieve a goal.

Each of these stages suggests natural design questions: How Easily Can One:

- Determine The Function of the Device? (Goal)
- Tell What Actions Are Possible? (Plan)
- Determine Mapping from Intention to Physical Movement? (Specify)
- Perform the Action? (Perform)
- Tell What State the System is In? (Perceive)
- Determine Mapping from System State to Interpretation? (Interpret)
- Tell if System is in Desired State? (Compare)

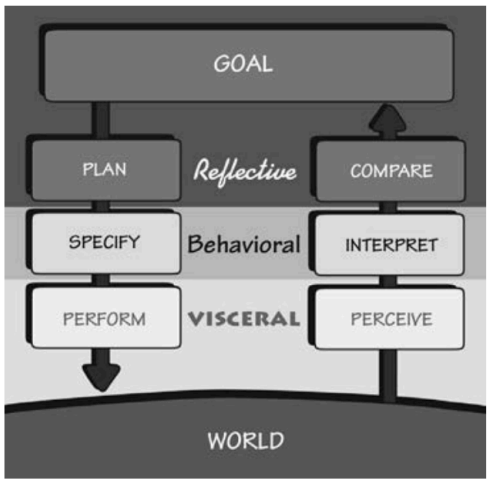


Figure 2.4 from @Norman2013Design.

David Marr, 2010, VISION: A Computational Investigation into the Human Representation and Processing of Visual Information, The MIT Press, Cambridge, Massachusetts, London, England @Marr:1982:VCI:1095712.

This book is a compelling story about the development of our understanding of how human vision works. Marr advocates an approach that separates concerns

> "For the subject of vision, there is no single equation or view that explains everything. Each problem has to be addressed from several points of view-as a problem in representing information, as a computation capable of deriving that representation, and as a problem in the architecture of a computer capable of carrying out both things quickly and reliably."

The current summary is limited to the introductory section, "Introduction and Philosophical Preliminaries", which advocates a mindset and approach for the more technical parts of the work which follow.

Marr describes experimental work which gives key insights into cognitive processes associated with vision. He cites Bela Jules work (1960) that led to random dot stereograms, which firmly established that 3D perception is firmly associated solely by the stereo disparity between matching elements in the images presented to each eye.

He also follows the development of electrophysiology by Adrian (1928), and successors who established

> "... the new view that questions of psychological interest could be illuminated and perhaps even explained by neurophysiological experiments."

By the 1970s this approach seemed to peter out.

> “The key observation is that neurophysiology and psychophysics have as their business to describe the behavior of cells or of subjects but not to explain such behavior.”

Marr turns to the approach of actually trying to implement the described behaviors. This leads to an algorithmic approach, including questions of efficiency and correctness.

The concepts of Representation and Process form a Computational Model of a system.

> “If the first of our levels specifies what and why, this second level specifies how. ... There are three important points here. First, there is usually a wide choice of representation. Second, the choice of algorithm often depends rather critically on the particular representation that is employed. And third, even for a given fixed representation, there are often several possible algorithms for carrying out the same process.”

Marr seeks to understand on 3 levels: The computational theory, the representation and algorithm, and finally the physical implementation of the process. He outlines a framework for understanding the derivation of shape in this way: The image, the primal sketch, the 2 1/2D sketch, and finally a 3D model representation.

The rest of the book covers these in detail with references to important experiments that support this understanding.

Colin Ware, 2012, Information Visualization: Perception for Design, Morgan Kaufmann Publishers, San Francisco, CA @ware2012information.

This book provides a large set of information design rules with supporting research. I have used it mainly as a reference work. I read through the index of design rules (Appendix D), referring back to the text for unfamiliar terminology. I look forward to reading the book cover-to-cover.

This book introduced me to the concept of geon theory, which proposes a hierarchical set of processing stages leading to object recognition. The top layer is a set of 3D primitives such as cones, cylinders, and boxes called *geons*. I am particularly intrigued by Biederman’s geon primitives used as an alternative to Unified Modeling Language (UML).

Here are a few of the guidelines and how I’ve applied them to SKOL.

> “[G1.1] Design graphic representations of data taking into account human sensory capabilities in such a way that important data elements and data patterns can be quickly perceived.”

I’ve tried to apply this by putting the most important controls at the top of the user interface.

> “[G6.1] Place symbols and glyphs representing related information together.”

Buttons that operate on fields are adjacent to the fields they apply to.

> "[G6.5] Consider putting related information inside a closed contour..."

Search results are delimited by a rounded box.

Andrew Csinger, 1992, The Psychology of Visualization, Department of Computer Science, University of British Columbia @Csinger1992Psychology.

> "This document is a review of the literature of three related areas: psychophysical vision research, automatic display generation, and multi-dimensional data visualization. Common threads are explored, and a model of the visualization process is proposed which integrates aspects of these three areas."

Csinger suggests a common vocabulary for understanding a range of authors who use different terminology, which some justifications for these choices.
The psychophysical review covers four authors, Pomerantz, Treisman, Enns, and Montalvo. Csinger describes the experiments they have used to determine various aspects of human visual perception, and some consequences of these experiments for data visualization.

Csinger's account of the Automatic Graph Generation literature starts off with the "highly idiosyncratic" author Bertin whose work lacks scientific rigor but offers a wealth of practical experience.

Mackinlay seems to offer more rigor, still empirical, but with plausible rationale.

> "He conjectures a theory of human perception that ranks the difficulty of perceptual tasks associated with the interpretation of presentations; a presentation composed of tasks which are perceptually 'easy' is more effective than one which is composed of 'harder' tasks."

Csinger does not hesitate to extend the work of various authors on multi-dimensional data visualization covering authors Pickett, Grinstein, Rogowitz et al., Enns, and others. He finishes the account of multi-dimensional data with some discussion of color including Ware and Beatty, and Murch.

Csinger offers a general model of visualization:

> "This model is intended as a general, covering model for a broad class of visualization activities, subsuming all the approaches discussed thus far in this paper."

The closing remark of the paper is a good summary of Csinger's aspiration:

> "While I think that it is premature to infer neurophysiological structure from the psychophysical studies alluded to, I see much to be gained by a practical marriage of visualization practitioners and psychophysicists."

This paper includes an extensive bibliography covering mainly the 1980's through 1991, though there are two papers from the 1970's and one from 1935 (Zipf of Zipf's Law).

Bibliography

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